

Topic 11: Advanced Techniques

11.1 Tree of Thought (ToT)

- Tree of Thought extends chain-of-thought by exploring multiple reasoning paths simultaneously
- The model generates several candidate 'thoughts', evaluates each, and pursues the most promising
- Useful for: creative problem solving, planning tasks, puzzles with multiple valid approaches
- Implementation: prompt the model to 'Generate 3 different approaches, evaluate each, then choose the best'
- More expensive than CoT but significantly better on tasks requiring search and backtracking

Problem: [describe the problem]

Step 1: Generate 3 different high-level approaches to solve this problem.
For each approach, briefly explain the reasoning.

Step 2: Evaluate each approach on: feasibility, time required, and risk.
Score each 1-10.

Step 3: Select the best approach and implement a detailed solution.

11.2 ReAct Prompting (Reason + Act)

- ReAct combines reasoning traces with action steps -- the model thinks aloud and decides what to do next
- Pattern: Thought -> Action -> Observation -> Thought -> Action -> ...
- Used in AI agents that can call tools (search, calculator, API calls)
- The 'Thought' step makes the model's reasoning transparent and debuggable

Thought: I need to find the current price of gold.
Action: search('gold price today USD')
Observation: Gold is trading at \$2,310 per troy ounce.

Thought: Now I need to convert to grams (1 troy oz = 31.1g).
Action: calculator('2310 / 31.1')
Observation: 74.27

Thought: I have the price per gram.
Final Answer: Gold costs approximately \$74.27 per gram today.

11.3 Generated Knowledge Prompting

- Ask the model to generate relevant background knowledge FIRST, then use it to answer the question
- Reduces errors by surfacing relevant facts before reasoning
- Two-step: Step 1 -- 'List key facts about X'; Step 2 -- 'Using these facts, answer Y'

Step 1:

List 5 important facts about photosynthesis relevant to why plants appear green.

Step 2:

Using the facts you just listed, explain in simple terms why leaves are green and not another colour.

11.4 Maieutic Prompting

- Ask the model to explain its reasoning step by step, then ask it to verify each step
 - Based on Socratic method -- expose and correct faulty assumptions
 - Particularly useful for detecting hallucinations in complex answers
 - Prompt: 'Explain your reasoning for each claim in your answer. Then verify each claim is correct.'
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